STEAMROOT · DESIGN DECISION RECORD

SteamRoot TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	STR-DDR-001	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/decisions/0001-trl2-review-decisions.md @ 276b380		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 8 and 9 decided by Amish on 2026-10-02; item 7 gate confirmed (STR-DEC-001, items 1 to 3)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the decisions Amish made on the TRL 2 review items, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 6); items 10 and 11 decided in STR-DDR-002; item 7 is an open prerequisite; items 8 and 9 remained proposed at this record and were decided by Amish on 2026-10-02 (STR-DEC-001, items 2 and 3): "i approve your recommendations for all 555 open decisions."

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-24) listed six items as "Proposed, awaiting Amish", each with a recommendation. On 2026-09-25 Amish reviewed the review points for every repo in the portfolio and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." This record lists what that instruction decides for SteamRoot and what it leaves open. TRL 4 work is on hold by the same instruction.

> **Safety:** Several of these decisions are safety architecture choices (open vent, relief valve, coil material). They are decisions about the concept only. They do not replace the written ruling from the local boiler authority, which is still an open prerequisite (item 7).

Options considered

The options for each item are in the TRL 2 review note and in `docs/02-concept.md` (STR-PRC-001 v0.2). Table 1 repeats the recommendation that was chosen.

Decision

Table 1. Items decided by Amish on 2026-09-25.

#	Item	Decision
1	Pressure architecture	Decided by Amish, 2026-09-25: go with recommendation. Open-vented steam side with a water-seal vent as the primary pressure limit, a certified relief valve as the second line, and no isolating valve anywhere on the steam side.
2	Coil type and material	Decided by Amish, 2026-09-25: go with recommendation. Single monotube coil in 25.4 mm OD 316 stainless tube (not parallel 12.7 mm circuits, not soft copper).
3	Treatment rate option	Decided by Amish, 2026-09-25: go with recommendation. Hood steaming with two hoods used alternately.
4	Feed control	Decided by Amish, 2026-09-25: go with recommendation. Fixed pump rate with a coil outlet temperature alarm (and the tank low-level alarm proposed with it at TRL 2).
5	Budget	Decided by Amish, 2026-09-25: go with recommendation (a). `budget_usd` raised from $1,500 to $1,800 to allow about 20 % contingency.
6	First users and first jurisdiction	Decided by Amish, 2026-09-25: go with recommendation. Involve one market garden and one nursery first; design the regulatory check to Amish's home jurisdiction first.

The TRL 2 review also recommended a written ruling from the local boiler authority before any TRL 3 detail work. Amish approved that recommendation with the rest. The ruling cannot be obtained in a documentation session, so the TRL 3 work was limited to paper calculations, a massing-plus model and a preliminary drawing, and the ruling is carried as an open prerequisite (item 7).

Table 2. Items that remain open.

#	Item	Status
7	Written ruling from the local boiler authority on an open-vented, fired monotube coil	Approved as a prerequisite; not yet obtained. Blocks detail design and any build. Confirmed by Amish on 2026-10-02 (STR-DEC-001, item 1): the design is reviewed against the ruling and anything stricter is adopted.
8	Which jurisdiction is "home" and which authority to ask	Decided by Amish, 2026-10-02 (STR-DEC-001, item 2): Texas; the ruling is asked of the Texas Department of Licensing and Regulation's boiler program.
9	The named market garden and nursery (first co-design partners)	Decided by Amish, 2026-10-02 (STR-DEC-001, item 3): one market garden and one nursery within driving distance of Irving that already steam, solarize or chemically treat soil, found through Texas A&M AgriLife Extension; none is named or agreed yet.
10	Wood only, no liquid fuel backup (design choice in STR-PRC-001 not listed in the review)	Decided by Amish, 2026-09-25: go with recommendation. Wood only. See STR-DDR-002.
11	New items raised by the TRL 3 calculation (CAL-001): requirement gaps on R2, R4, R5, R6, R11 and R12, relief valve wording for R9, and the lining and hose-coupling choices	Decided by Amish, 2026-09-25: go with recommendation. See STR-DDR-002 for each item and what changed.

Consequences

- `project.yaml` : `budget_usd` is 1800.
- `docs/03-requirements.md` (STR-REQ-001 v0.3): R2 is stated for two hoods; R12 is $1,800.
- `docs/02-concept.md` (STR-PRC-001 v0.3): the open vent, the stainless monotube coil, two hoods and the fixed-rate feed with alarms are recorded as decided.
- `docs/01-problem.md` (STR-PRB-001 v0.3): first users and first jurisdiction are recorded as decided; the named partners and the named jurisdiction stay open.
- `bom/bom.csv` : a second hood and the feed and coil alarms are added.
- The TRL 3 calculation (STR-CAL-001) shows that the decided configuration misses several requirements. Those results do not reverse any decision here; they are new items for Amish.